

Complex and Majorana covariance matrices with an internal index

1 Setup

Consider N complex fermionic modes with an additional internal label $\mu \in \{+, -\}$, satisfying $\{c_{i,\mu}, c_{j,\nu}^\dagger\} = \delta_{ij}\delta_{\mu\nu}$. Define the *complex covariance*

$$G_{ij,\mu\nu} = \langle [c_{i,\mu}^\dagger, c_{j,\nu}] \rangle = 2\langle c_{i,\mu}^\dagger c_{j,\nu} \rangle - \delta_{ij}\delta_{\mu\nu}, \quad i, j = 1, \dots, N, \quad (1)$$

and let $\mathbf{G}_{\mu\nu}$ denote the $N \times N$ matrix with entries $G_{ij,\mu\nu}$. The full $\mathbf{G}$ (lumping (i, μ) into one index) is Hermitian:

$$(\mathbf{G}_{\mu\nu})^\dagger = \mathbf{G}_{\nu\mu}, \quad G_{ij,\mu\nu}^* = G_{ji,\nu\mu}. \quad (2)$$

Switch to Majoranas via

$$c_{i,\mu} = \frac{1}{2}(\gamma_{2i-1,\mu} + i\gamma_{2i,\mu}), \quad \{\gamma_{a,\mu}, \gamma_{b,\nu}\} = 2\delta_{ab}\delta_{\mu\nu}, \quad (3)$$

so that $\gamma_{2i-1,\mu} = c_{i,\mu} + c_{i,\mu}^\dagger$, $\gamma_{2i,\mu} = -i(c_{i,\mu} - c_{i,\mu}^\dagger)$. The Majorana covariance is

$$\Gamma_{ab,\mu\nu} = \frac{i}{2}\langle [\gamma_{a,\mu}, \gamma_{b,\nu}] \rangle, \quad a, b = 1, \dots, 2N, \quad (4)$$

with $\mathbf{\Gamma}_{\mu\nu}$ the corresponding $2N \times 2N$ matrix. The full $\mathbf{\Gamma}$ is real antisymmetric.

Scope. The Bogoliubov rotation $c \rightarrow \gamma$ does not mix the internal index, so $\mathbf{\Gamma}_{\mu\nu}$ depends only on the *same* block $\mathbf{G}_{\mu\nu}$. The complex covariance G however encodes only the normal two-point function $\langle c^\dagger c \rangle$; the Majorana object additionally sees the anomalous correlators $\langle cc \rangle, \langle c^\dagger c^\dagger \rangle$. The problem "express $\mathbf{\Gamma}$ in terms of $\mathbf{G}$ " is therefore only well-posed when the state is $U(1)$ -symmetric, so that

$$\langle c_{i,\mu} c_{j,\nu} \rangle = 0, \quad \langle c_{i,\mu}^\dagger c_{j,\nu}^\dagger \rangle = 0. \quad (5)$$

With pairing, an independent anomalous block $F_{ij,\mu\nu} = \langle c_{i,\mu} c_{j,\nu} \rangle$ must be supplied. We assume (5) from here on.

2 Main result

Using (2), $\langle [c_{i,\mu}, c_{j,\nu}^\dagger] \rangle = -G_{ji,\nu\mu} = -G_{ij,\mu\nu}^*$. Expanding the four combinations of $(\gamma_{\text{odd}}, \gamma_{\text{even}})$ and dropping the anomalous terms:

$$\langle [\gamma_{2i-1,\mu}, \gamma_{2j-1,\nu}] \rangle = \langle [\gamma_{2i,\mu}, \gamma_{2j,\nu}] \rangle = G_{ij,\mu\nu} - G_{ij,\mu\nu}^* = 2i \Im G_{ij,\mu\nu}, \quad (6)$$

$$\langle [\gamma_{2i-1,\mu}, \gamma_{2j,\nu}] \rangle = -\langle [\gamma_{2i,\mu}, \gamma_{2j-1,\nu}] \rangle = -2i \Re G_{ij,\mu\nu}. \quad (7)$$

Multiplying by $i/2$ gives the inner 2×2 block at mode pair (i, j) :

$$\begin{pmatrix} \Gamma_{2i-1,2j-1} & \Gamma_{2i-1,2j} \\ \Gamma_{2i,2j-1} & \Gamma_{2i,2j} \end{pmatrix}_{\mu\nu} = \begin{pmatrix} -\Im G_{ij,\mu\nu} & \Re G_{ij,\mu\nu} \\ -\Re G_{ij,\mu\nu} & -\Im G_{ij,\mu\nu} \end{pmatrix}. \quad (8)$$

In the natural interleaved ordering $(\gamma_1, \gamma_2, \gamma_3, \dots)$, this is a Kronecker product with the $N \times N$ mode factor on the outside and the 2×2 "odd/even" factor on the inside:

$$\boxed{\Gamma_{\mu\nu} = -\Im \mathbf{G}_{\mu\nu} \otimes \mathbb{I}_2 + \Re \mathbf{G}_{\mu\nu} \otimes J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},} \quad (9)$$

or equivalently

$$\Gamma_{\mu\nu} = \frac{i}{2} [\mathbf{G}_{\mu\nu} \otimes (\mathbb{I} + \sigma_y) - \mathbf{G}_{\mu\nu}^* \otimes (\mathbb{I} - \sigma_y)]. \quad (10)$$

3 Consistency checks

Reality. $\Re \mathbf{G}, \Im \mathbf{G}, \mathbb{I}_2, J$ are all real, so $\Gamma_{\mu\nu}$ is real.

Antisymmetry of the full Γ . We need $(\Gamma_{\nu\mu})^T = -\Gamma_{\mu\nu}$. Using (2) in the form $\mathbf{G}_{\nu\mu}^T = \mathbf{G}_{\mu\nu}^*$, hence $\Re(\mathbf{G}_{\nu\mu}^T) = \Re \mathbf{G}_{\mu\nu}$ and $\Im(\mathbf{G}_{\nu\mu}^T) = -\Im \mathbf{G}_{\mu\nu}$, together with $J^T = -J$:

$$(\Gamma_{\nu\mu})^T = -\Im \mathbf{G}_{\mu\nu} \otimes \mathbb{I}_2 - \Re \mathbf{G}_{\mu\nu} \otimes J = -\Gamma_{\mu\nu}. \quad \checkmark$$

Single mode. For $N = 1$ and a single species, $\langle c^\dagger c \rangle = n$ gives $G = 2n - 1 \in \mathbb{R}$, so (9) yields $\Gamma_{11} = \Gamma_{22} = 0$ and $\Gamma_{12} = -\Gamma_{21} = 2n - 1$, which agrees with $\frac{i}{2} \langle [\gamma_1, \gamma_2] \rangle$ computed directly.

Purity. For a $U(1)$ -symmetric pure Gaussian state the normal covariance satisfies $\mathbf{G}^2 = \mathbb{I}$ (the shifted correlation matrix is $\pm \mathbb{I}$ on filled/empty orbitals). Together with $J^2 = -\mathbb{I}$, equation (9) implies $\Gamma^2 = -\mathbb{I}_{2N}$, as required for a pure Majorana Gaussian state.

4 Interpretation

Identify $\mathbb{C}^{N \times N} \hookrightarrow \mathbb{R}^{2N \times 2N}$ via the real representation $\Phi(a + ib) = a \mathbb{I}_2 - b J$, which satisfies $\Phi(z_1 z_2) = \Phi(z_1) \Phi(z_2)$ (using $J^2 = -\mathbb{I}$). A short calculation shows

$$\Gamma_{\mu\nu} = \Phi(-i \mathbf{G}_{\mu\nu}^*). \quad (11)$$

The factor $-i$ is the complex structure that converts the Hermitian $\mathbf{G}$ into the antisymmetric Γ . This is the same doubling that makes Gaussian channels act linearly on Majorana covariances, $\Gamma \mapsto T \Gamma T^\top$, while acting on $\mathbf{G}$ only by a Möbius / Schur-complement map.